Project Proposal

Project: Eduvance AI

Objective:
Design, implement, and evaluate an integrated agentic AI-assisted learning platform that transforms a bounded set of training documents into structured, personalized, and source-traceable learning experiences.

Description:
Eduvance AI is a Digital Egypt Pioneers Initiative (DEPI) capstone project within the AI & Data Science - Agentic AI and Generative AI System Developer track. It aims to process text-based PDFs (up to three files / 50 pages per course), extract concepts and source references, propose an editable curriculum (approximately three modules and six lessons), generate narrated slide-based lessons, support source-grounded RAG tutoring, create assessments, track learning events, recommend revisions, and suggest a bounded practical project. These are prototype objectives, not claims that every feature is already complete.

Tools & Design:
Layered application design with a central workflow controller coordinating the Content, Course Designer, Video, Tutor, Assessment and Project Mentor agents, alongside deterministic progress and scoring. Typed agent contracts, source anchors (document/page/chunk), structured persistence, course review and approval, state transitions, and bounded retries support integration and traceability.

Technologies to Use:
Python, FastAPI, Pydantic v2, SQLAlchemy 2.0, Alembic and SQLite for local development; PostgreSQL and FAISS/pgvector as design options for persistence and retrieval. Planned interface: React with Tailwind CSS. Supporting tooling includes PDF parsing, slide rendering, text-to-speech, captions and automated testing. Existing AI models/services are integrated rather than trained from scratch.

Team Members & Responsibilities:

1- Mostafa Mahmoud Saleh GadAllaha - Agentic AI Orchestration & Integration Lead; planning, system architecture, workflows, agent contracts, backend integration and progress logic (Member 1).
2- Hana Amr Mahmood Ahmed - RAG & Knowledge Intelligence Engineer; document extraction, concepts, course generation, source-grounded tutor and grounding evaluation (Member 2).
3- salma rafat anwar elhity - Generative AI Content & Assessment Engineer; generated lesson media, question generation, practical rubrics and Project Mentor (Member 3).
4- Assem Taher Abouzaid Elgamal - AI Evaluation & Personalization Engineer; user journeys, interface implementation, integration-test coordination, learner-facing feedback and usability evaluation (Member 4).

Week 1:

- Member 1 (Mostafa): confirm DEPI capstone requirements, review expectations and marking criteria (WBS 1.1).
- Member 1 (Mostafa): define the problem statement and project objectives (WBS 1.2).
- Member 1 (Mostafa): establish the agreed prototype scope and constraints (WBS 1.2).
- Member 1 (Mostafa): prepare the project proposal (WBS 1.2).
- Member 1 (Mostafa): align the initial backlog and team responsibilities with the scope (initiation phase).

Deliverables:

- Initial project proposal and problem statement.
- Defined project objectives and bounded scope.
- Project charter and initial backlog.
- DEPI requirements and review expectations recorded.

Week 2:

- Member 2 (Hana): begin reviewing related educational AI research and systems (WBS 1.3).
- Member 2 (Hana): develop the related-systems comparison table (WBS 1.3).
- Member 2 (Hana): identify evaluation questions and candidate metrics (WBS 1.4).
- Member 3 (salma): assess feasibility of lesson scripts, narration and captions (WBS 2.3).
- Member 3 (salma): examine video rendering time and operating-cost constraints (WBS 2.3).
- Member 1 (Mostafa): begin consolidating the project report and evidence (WBS 7.1).

Deliverables:

- Literature review draft and research notes.
- Related-systems comparison in progress.
- Initial evaluation questions, metrics and candidate dataset.
- Preliminary video-feasibility observations.

Week 3:

- Member 2 (Hana): continue research review and comparative findings (WBS 1.3).
- Member 2 (Hana): define evaluation datasets, measures and baseline methodology (WBS 1.4).
- Member 3 (salma): complete video narration, captions and rendering feasibility work (WBS 2.3).
- Member 1 (Mostafa): begin system architecture and agent integration contracts (WBS 2.1).
- Member 4 (Assem): begin user journeys, wireframes and interface component design (WBS 2.2).
- Member 1 (Mostafa): begin environment, database and storage setup (WBS 3.1).

Deliverables:

- Research and feasibility findings for extraction, retrieval and video generation.
- Initial evaluation dataset and protocol.
- Architecture and technology-design decisions in progress.
- Early requirements and user-journey drafts.

Week 4:

- Member 1 (Mostafa): continue and finalize system architecture and data model (WBS 2.1).
- Member 1 (Mostafa): define typed agent interfaces and integration contracts (WBS 2.1).
- Member 4 (Assem): finalize user journeys, wireframes and UI component specifications (WBS 2.2).
- Member 1 (Mostafa): continue repository, environment, database and storage setup (WBS 3.1).
- Member 2 (Hana): begin document-upload validation and extraction work (WBS 3.2).
- Member 2 (Hana): begin designing source-linked text and reference outputs (WBS 3.2).

Deliverables:

- Requirements specification and use cases.
- System architecture diagram and agent integration contracts.
- Database design and relational model.
- User journeys, wireframes and interface prototype design.